CHRIS FRISINA

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SUMMARY

Engineering leader and full-stack architect who has led teams of 3 to 30 people across logistics operations and software. I design secure multi-tenant platforms, run red, blue, and purple team security work, and build agentic development systems, grounded in human-centered design, consulting, and logistics.

PROFESSIONAL EXPERIENCE

DEC 2025–PRESENT

Lead Engineer, MyFleet.tech.

Multi-tenant SaaS platform that connects fleets with their maintenance shops: assets, inspections, preventive maintenance, work orders, estimates, and invoicing.

- Architected a multi-tenant SaaS platform with per-customer data isolation, custom subdomains, and white-label branding.
- Built subscription billing (plans, add-ons, usage metering, proration) and integrations with accounting and telematics providers.
- Delivered inspection, kiosk, vendor portal, SMS, and AI support features for both field and office users.
- Established a testing culture of 16,000+ automated tests running in isolated containers, with coverage reporting in CI.
- Created reproducible, resettable environments: one-command reset and seeding, self-resetting demo accounts, and per-branch development stacks.
- Automated zero-downtime deployments across two cloud providers, with centralized secrets management and security hardening.
- Stack: Python, Django, DRF, React, PostgreSQL, Celery, Redis, Docker, Stripe, AWS, Twilio, OpenBao, GitHub Actions.

APR 2016–
PRESENT

Full Stack Engineer, Master Specialists, Charlotte, NC.

Fleet maintenance, inspection, and DOT compliance platform for delivery and logistics fleets. Django / React platform used daily by mechanics, dispatch, fleet managers, and billing.

- Designed and led a multi-phase migration to a finite state machine workflow engine with full audit history; released the engine as open source (django-fsm-rx on PyPI).
- Delivered zero-downtime blue-green deployments, containerization, and modern dependency management.
- Centralized secrets management with encryption, failover, and key rotation; restricted file storage to time-limited access.
- Automated billing and finance: invoicing, tax filing, ACH payments, and bank-feed expense reconciliation with receipt matching.
- Integrated AWS and S3 to resolve email queue bottlenecks and support asynchronous processes.
- Built dispatch and service workflows with role-based views, customer-facing status pages, and automated vendor notifications.
- Built internal operations tooling: payroll review, team dashboards, and scheduled reporting.
- Designed partner APIs with event callbacks for cross-platform integrations.
- Built APIs, reusable components, and reporting across the platform; refactored legacy code into maintainable services.
- Optimized database queries on high-traffic paths and expanded unit and integration testing to improve uptime.
- Automated regulatory, procedural, and cost-saving logistics processes; led client design and support meetings.
- Stack: Python, Django, DRF, React, TypeScript, PostgreSQL, Celery, RabbitMQ, Twilio, Square, Plaid, AWS, OpenBao, Docker, GitHub Actions, Linux.

JAN 2013–
PRESENT

Graduate Researcher, The Sound of Fractions, Virginia Tech, Blacksburg, VA. Try it live.

Designed, built, and ran a study of a full stack web application and curriculum that lets students who struggle with fractions tap rhythms, which audio processing turns into graphical fraction representations. Built a data viewer for cross-discipline analysis of the results.

- Funded by a $25K student research grant. Featured on Virginia Public Radio.
- Stack: Node, Backbone, D3, HTML5 Audio, Parse, Bootstrap, Ubuntu.

DEC 2012–
DEC 2015

Graduate Researcher, Virginia Tech, Blacksburg, VA.

Researched how to help entering undergraduates succeed by making data meaningful to them; built a customizable learning-analytics dashboard for students, teachers, and administrators (proof of concept).

FEB 2006–
DEC 2014

Implementation Manager, FedEx Ground.

Helped develop and implement an automated sortation and dispatch system for final-stage package delivery. Reduced costs 5% and raised accuracy 4%; security and Six Sigma improvements saved more than $10M.

- Led teams of 3 to 30 people across service, quality, and implementation roles.
- Disclosed a $115M security issue to the C-suite in 2008, testified on it, and led the fixes and the security education that followed.
- Contractor Corporate Liaison, Jul 2010–Dec 2014.
- Vision Implementation Manager, Feb 2010–Dec 2011.
- Southern Division Quality Initiative Manager, Feb 2009–Jan 2011.
- Senior Service Manager, Jan 2007–Jan 2009.
- Service Manager, Apr 2006–Dec 2006. Package Handler, Mar 2006–May 2006.

MAY 2002–
FEB 2006

Grocery Manager, Harris Teeter.

Managed grocery departments for several area stores; designed an improved inventory system.

SECURITY

2008

Responsible disclosure, FedEx Ground.

- Identified and responsibly disclosed a $115M security issue, escalating it directly to the C-suite.
- Testified on the findings and worked with leadership to fix the underlying issues.
- Spearheaded the education programs and services that followed, focused on safe and secure operations.

2025–
PRESENT

Penetration testing: red, blue, and purple team assessments of production at Master Specialists and MyFleet.tech.

- Ran full-stack assessments covering perimeter, auth, data, secrets, containers, host, supply chain, and CI/CD. Red team exploited attack paths live against production, blue team hardened detection, purple team verified every claim by execution rather than config review.
- Closed findings including a container-to-cloud-metadata privilege escalation path (to an administrator cloud role), a non-expiring root secrets token replaced with a scoped one, and missing secrets audit logging.
- Code-level red team review of MyFleet.tech (15 findings, 2 critical); fixed authorization bypasses, enforced tenant scoping across views and APIs, and run a recurring tenant isolation audit against production.

2017, 2026

Digital forensics and incident response (DFIR) on a compromised multi-site hosting account.

- Triaged and remediated a long-running compromise across 12 sites: found and quarantined about 370 artifacts, including web shells, obfuscated PHP loaders hidden in JavaScript libraries, payloads disguised as CSS and image files, search-engine spam redirects, a fake WordPress plugin and theme, and two backdoor administrator accounts.
- Built signature and heuristic scanners for the malware families, verified WordPress core and plugins against official checksums, and surgically removed injected code from legitimate files, keeping originals for evidence and rollback.
- Linked the 2026 infection to artifacts left from a 2017 compromise, worked with the host's abuse team in 2017, and hardened the account afterward (key-only SSH, credential rotation, patched CMS and plugins).

INFRASTRUCTURE

- Secrets in OpenBao (Vault) with transit encryption, a standby node, rapid key rotation, and audit logging.
- Private object storage behind presigned URLs; forced HTTPS, HSTS, CSP, MFA, rate limiting, Cloudflare Turnstile, and bot tarpits.
- Email authentication with SPF, DKIM, DMARC, and MTA-STS.
- Supply chain: secret scanning in pre-commit and CI, Dependabot alerts gated by a required test suite.

ARCHITECTURE & AGENTIC SYSTEMS

ARCHITECTURE

- **MyFleet.tech:** multi-tenant SaaS with schema-per-tenant PostgreSQL, subdomain routing, and white-label branding; a two-server production design separating the app (Django, PostgreSQL, Redis, Celery, OpenBao, Nginx) from observability (Prometheus, Grafana, Loki, GlitchTip, Alertmanager, Uptime Kuma); blue-green Docker deploys to AWS and Hetzner.
- **Master Specialists:** moved production from a single virtualenv to Docker containers sized to measured memory, with zero-downtime blue-green deploys from GitHub Actions, an OpenBao secrets tier with a standby node, and a finite state machine core for regulatory record tracking.
- **Sound of Fractions:** revived a 2015 research app as static hosting plus a Cloudflare Worker and D1 database for interaction logging.

AGENTIC ENGINEERING

- Designed a multi-agent development system with Claude Code: a planner hands each task to a fresh subagent, then reviews its work before the next task starts.
- Built domain-specialist agents per codebase (data, background jobs, APIs, security, deployment, testing, front end, email) plus reusable skills such as red-team and blue-team review and test-first development.
- Parallel agents run in isolated git worktrees, each with its own Docker stack (containers, ports, database volume), so several features progress at once without collisions.
- An always-on agent server on Hetzner with its own repo, Docker stack, and database copy keeps working while my laptop is off; reachable from phone, web, or SSH, with mobile alerts.
- Agents share a version-controlled issue tracker synced between machines, so work and context carry across sessions and hosts.
- Guardrails: agents commit but never push to production branches, are blocked from destructive database actions, and every pull request gets automated tests and AI code review in CI.
- Shipped an in-app AI support assistant with semantic search that escalates to staff.

EDUCATION

2013–2016 **M.S., Computer Science (Human Computer Interaction)**, Virginia Tech. GPA 3.7.

Thesis: The Sound of Fractions. Coursework: CSCW & Social Networks, Human Centered Methods, Programming Languages, Digital Libraries, Usability Engineering, Computational Thinking, Research Methods.

2003–2007 **B.A., Communication (Female Gendered Communication)**, University of North Carolina at Charlotte.

PUBLICATIONS

APR 2016 The Sound of Fractions. Master's thesis, Virginia Tech. Teaching fractions through music with an online application and curriculum.

MAY 2014 Knowledge Building and Sharing: A meta-model for Guided Research, Learning, and Application. VTechWorks.

2014 RaBit EscAPE: a board game for computational thinking. Interaction Design and Children (IDC).

SKILLS

Areas: Full stack web development, UI/UX, data visualization, big data, open data, systems analysis, heuristic evaluation, participatory and human centered design, time and method engineering, pedagogy.

Languages & tools: Python, Django, JavaScript, TypeScript, React, Node, D3, SQL / PostgreSQL, Celery, RabbitMQ, AWS, HTML/CSS, Bootstrap, R, Git, Linux.

Certifications: Six Sigma, NERC, CIP, SCADA, OSHA, DOT, ISO.

Spoken: English (native), Spanish (conversational), German (beginner).

AWARDS

JAN 2013	Student Research Grant ($25K), Virginia Tech.
JUN 2009	Most Improved Service Manager, FedEx.
FEB 2003	Eagle Scout, Boy Scouts of America.

VOLUNTEER WORK

APR 2016–PRESENT	**Planning Director**, IUPUI Relief Housing. Housing for adults recovering from trauma.
AUG 2014–2016	**Social Chair**, Virginia Tech Computer Science Grad Council.
FEB 2012–FEB 2013	**Committee Chair**, Ecumenical Think Tank. Developed opportunities to grow young adult participation in community small groups; allocated $100K among them.
JAN 2001–PRESENT	**Team Member**, Habitat for Humanity. Framing and roofing; supply chain support for Bahamas Habitat (1000+ families); Gulf disaster relief.

INTERESTS

Outdoors (camping, hiking, whitewater kayaking, mountain biking, snowboarding) · Cooking (large scale 100+, fundraisers) · Woodworking (carpentry, turning, rare woods) · Neon and custom artistic signs.

REFERENCES

Michael Stewart, tgm@vt.edu · Melissa Avery-Wier, averymd@irrsinn.net